

Jayson Mendoza

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EDUCATION

The University of Texas at Arlington

Expected Date: May 2028

B.S. in Mechanical Engineering | Minor in Business Administration

Member: Society of Hispanic Professional Engineers (SHPE), inSTEM, The Vertical Flight Society

EXPERIENCE

Research Assistant | AIM³ Lab (Artificial Intelligence and Machine Learning for Multiscale Modeling)

University of Texas at Arlington Research Institute— Advisor: Xin Liu, Ph.D.

Feb 2026 – Present

- Adapted open source Air Force Research Laboratory Python finite element code to investigate material behavior and structural damage through differentiable and multiscale modeling.
- Built and simulated structures with hundreds of finite elements, defining mesh geometry, material properties, loading conditions, constraints, and structural boundary conditions.
- Trained neural networks to predict material properties and structural damage from simulated finite element displacement data.
- Evaluated structural damage and stiffness degradation by comparing predicted and simulated mechanical responses, and generated convergence and damage-localization plots to quantify identification error.

PROJECTS

MATLAB 3D Finite Element Solver | MATLAB, Finite Element Method

May 2026 - Aug 2026

- Developed a MATLAB program for linear elastic 3D frame analysis using the direct stiffness method.
- Implemented beam and frame finite elements, global stiffness matrix assembly, boundary condition enforcement, and nodal displacement and support reaction calculations.
- Modeled materials, cross sections, supports, releases, and nodal, concentrated, and distributed loads for structural analysis.
- Computed internal member force distributions (axial, shear, bending, and torsion) and visualized structural geometry, finite element meshes, and deformed shapes in MATLAB.

Custom Quadcopter | SolidWorks, ANSYS, Finite Element Analysis

Jan 2026 - Aug 2026

- Designed a PETG quadcopter frame in SolidWorks and prepared CAD geometry for simulation in ANSYS Mechanical.
- Performed static structural FEA under motor thrust and gravitational loading to evaluate von Mises stress, elastic strain, and total deformation across the frame and drone arm.
- Developed and refined finite element meshes, including local body sizing and geometry simplification, to resolve complex lattice structures and improve mesh quality.

pH Monitoring System | Arduino, C++, CAD, Sensors

July 2025 - Oct 2025

- Designed and built a pH monitoring system using an Arduino and pH probe for real time water quality measurement.
- Calibrated the sensor using buffer solutions and wrote Arduino code to convert probe readings into accurate pH measurements.
- Fabricated a custom enclosure integrating electronics and peristaltic pumps with organized wiring for mechanical stability.

Tennis Ball Retrieval Device | SolidWorks, GD&T, Finite Element Analysis

Jan 2025 - May 2025

- Designed and built a robot using Arduino to autonomously capture a free falling tennis ball and transport it 6 feet into a tray within 30 seconds.
- Created custom parts in SolidWorks with GD&T compliant drawings; fabricated components via 3D printing.
- Performed motion simulation and stress analysis to verify structural performance.

TECHNICAL SKILLS

Mechanical Design & CAD: SolidWorks (CSWA), GD&T, Bill of Materials (BOM)

Analysis and Testing: ANSYS, Finite Element Analysis (FEA), Motion Simulation, Stress Analysis

Programming: Python(NumPy, SciPy, Matplotlib, JAX), MATLAB, Basic C++

Productivity: Microsoft 365 (Excel, Word, PowerPoint)